ENERGY SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military Action Extension Methodology

DBA-EN-SF: Energy Sector Parent Document

Governing Parent for DBA-ES, DBA-GAS, DBA-OIL,
and All Sub-Sector Instruments in the Energy Sector Series

Version 0.3 — DRAFT

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For review by engineers, regulators, asset owners, and policy analysts responsible for protecting energy sector critical infrastructure across the electric, natural gas, and petroleum sub-sectors.

Revision History

Version	Date	Author	Description
0.1	May 2026	T. Roxey	Initial working draft. Established Energy sector parent document architecture. Defined three sub-sector series: DBA-ES (Electric), DBA-GAS (Natural Gas), DBA-OIL (Petroleum). Established an inductive architecture principle. Named the gas-electric bidirectional cascade as the governing cross-sub-sector interdependency. Confirmed DBA-ES retains its own designation and is not renamed under a DBA-EN hierarchy.
0.2	June 2026	T. Roxey	Substantive revision to serve as the authoritative parent document for the new Energy Sector project. Incorporated the complete Energy sector taxonomy from Design Basis Energy Sector Mindmap v2 (June 2026). Added full GenCo taxonomy (fossil fuel, nuclear, renewable sub-types). Added TransCo infrastructure detail (voltage class, bus bar identifiers, ICCP, regulatory anchors). Added DisCo control systems detail (EMS, PMU, SCADA polling cycle). Added CntlCo completeness: RC tool suite (OATI, EPRI TagNet, Power World, RCIS), BA tool suite (RCIS, EMS, Security Constrained Unit Commitment), NERC CIP compliance registry (BCA, EACMS, PACS, PCA), primary and backup control center communications architecture. Added Oil and Natural Gas sub-sector taxonomy: crude oil (upstream, terminals, transport, storage), petroleum processing (refineries, terminals), natural gas (production, processing, transport, distribution, storage, LNG). Formalized gas-electric bidirectional cascade as a first-class cross-sub-sector postulated event category. Added sub-sector series inventory table with the current version states. Added dual-DB architecture acknowledgment per DBA-ES §3.4. Added Bright Line and Command Broker as cross-sector series requirements. Added series-wide naming convention—added references section.
0.3	June 2026	T. Roxey	Section 3.4 (Cross-Sector Consequence Elevation Rule) added. Establishes the governing coordination protocol for DBA-ES-GC-FC4 and DBA-GAS-FC1 at dual-scope compressor station facilities: consequence-perspective separation rule (each instrument analyzes from its own consequence chain); acceptance criteria assignment rule (each instrument's criteria govern its own consequence tier); and cross-sector consequence elevation rule (a DBA-EN energy asset that is the sole supply path for a Tier 2 or Tier 3 DBA-designated facility in any sector must carry tightened acceptance criteria for that supply relationship). Names the NERC CIP gap explicitly: CIP does not require cross-sector consequence elevation; §3.4 establishes this as a pre-regulatory design basis requirement. Adopts the single-path dependency criterion from FC4 §1.2 as the elevation trigger threshold. Documents the unidirectional current implementation and the intended bidirectional future state. OI-1 closed. Section 5.3 (Nuclear Sector Grid Dependency) expanded: NUC-001-4 (Nuclear Plant Interface Coordination) added as the NERC regulatory instrument governing the nuclear-grid operational interface; the relationship between NUC-001-4's NPIR mechanism and the DBA-EN design basis framework is stated explicitly. NUC-001-4

			added to the references section—updated DBA-GAS-FC1 to v1.5 and DBA-ES-GC-FC4 to v0.2 in the series inventory table.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the NIPP Energy sector. It is the parent instrument for three sub-sector series: DBA-ES (Electric sector), DBA-GAS (Natural Gas sector), and DBA-OIL (Petroleum sector). Each sub-sector series inherits the methodology and organizing principles established here and applies them to the specific facility types, operational characteristics, and regulatory contexts of its sector.

The Energy sector, as defined by the National Infrastructure Protection Plan (NIPP), encompasses the electric power system, natural gas production and delivery infrastructure, and petroleum production and refining infrastructure. DHS is the Sector-Specific Agency (SSA) for the Energy sector, with DOE, FERC, NERC, TSA, and PHMSA as the primary federal regulatory and operational partners depending on sub-sector and function.

This document applies the DB→DBA→DBA-MA three-layer analytical progression at the Energy sector level: a baseline Design Basis (DB) establishing the postulated conditions the sector's facilities must withstand; a Design Basis Accident (DBA) layer bounding the non-adversarial consequence envelope; and a military action extension (DBA-MA) applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to the sector's governing postulated events. The sector-level instruments establish this progression as a requirement; each sub-sector framework and facility-class instrument develops it in detail.

1.2 What This Document Does Not Do

This document does not supersede or modify any existing sub-sector framework or facility-class instrument. DBA-ES, DBA-GAS, and DBA-OIL are established series with their own governing documents; this parent establishes the common architecture they share and the cross-sector cascade analysis that spans their boundaries.

This document does not rename the sub-sector series. DBA-ES, DBA-GAS, and DBA-OIL retain their existing designations. The DBA-ES series retains its internal naming convention (DBA-ES-HY-FC1, DBA-ES-TX-FC2, etc.) without modification.

This document does not create new facility class designations. FC designations are the authority of their respective sub-sector frameworks. This document carries a series of inventory tables that reflect the current FC register state across all three sub-sectors.

1.3 Inductive Architecture Principle

The Energy sector series is built inductively: parent documents emerge from mature children, not the reverse. DBA-EN-SF v0.3 is drafted at this version because the three sub-sector frameworks have now reached sufficient maturity to reveal the common architecture they share. This document does not direct the sub-sector frameworks; it reflects and codifies what they have established.

Future revisions of DBA-EN-SF will be triggered by substantive developments in the sub-sector frameworks — new facility classes, new cross-sector cascade methodologies, or changes to the governing regulatory context — not by a top-down revision schedule.

1.4 Sub-Sector Series Inventory

Sub-Sector Series	Sector Framework	Current Version	SSA	Active FC Instruments
DBA-ES (Electric)	DBA-ES-Sector_Framework	v0.6 — DRAFT	DOE / FERC / NERC	DBA-ES-HY-FC1 (v0.4), DBA-ES-TX-FC2 (v0.1), DBA-ES-TX-FC3 (v0.1), DBA-ES-GC-FC4 (v0.2), DBA-ES-DI-FC5 (v0.1), DBA-ES-CntlCo (v0.1), DBA-DC (v1.5)
DBA-GAS (Natural Gas)	DBA-GAS-Sector_Framework	v1.4 — DRAFT	DOE / FERC / TSA / PHMSA	DBA-GAS-FC1 (v1.5), DBA-GAS-UC-LNG (v1.1), DBA-GAS-POL_Deterrence (v1.2)
DBA-OIL (Petroleum)	DBA-OIL-Sector_Framework	v1.3 — DRAFT	DOE / TSA / PHMSA / EPA	DBA-OIL-FC1 (v1.1)

Section 2 — Energy Sector Taxonomy

2.1 Organizing Framework

The Energy sector's facility taxonomy is organized into two primary branches corresponding to the NIPP Energy sector structure: Electricity and Oil and Natural Gas. The Electricity branch covers generation (GenCo), transmission (TransCo), distribution (DisCo), and the control-company function (CntlCo). The Oil and Natural Gas branch covers crude oil production, midstream, and petroleum processing; and natural gas production, transmission, distribution, LNG, and storage.

The taxonomy in this section is the authoritative reference for series-wide facility scoping decisions. When a new facility class is proposed for any sub-sector series, its placement — which sub-sector series it belongs to, which consequence chain governs its inclusion, and which regulatory anchor applies — is determined against this taxonomy. The governing principle is consequence-chain placement: a facility belongs in the sub-sector series whose primary consequence chain it affects most directly.

2.2 GenCo — Generation by Type

The generation taxonomy spans seven primary generation types. Nuclear generation is addressed in the DBA-MA nuclear series and is referenced here for completeness; its grid consequence chain (Zaporizhzhia: loss of Europe's largest nuclear plant as both a nuclear safety event and a grid reliability event) is addressed in the DBA-ES series and the DBA-MA Introduction.

Generation Type	Sub-Types	Primary Regulators	DBA-EN Series Treatment	FC Status
Fossil fuel	Coal, natural gas, oil-fired	TSA (security); NERC/FERC (reliability); State environmental (emissions)	Gas-fired generation is addressed through a gas-electric cascade in DBA-ES-GC-FC4 and DBA-GAS-FC1. Coal and oil-fired: deferred FC.	Partial (gas via FC4)
Nuclear	Large LWR (PWR, BWR), SMR, VVER	NRC (US); STUK (Finland); SNRIU (Ukraine); IAEA (international)	Addressed in the DBA-MA nuclear series. Grid consequence chain (Zaporizhzhia) addressed in DBA-ES and DBA-MA Introduction §6.3.	DBA-MA series
Hydropower	Large storage (>500 MW), run-of-river, pumped storage, tidal	FERC Part 12; NERC CIP-014; USCOLD; Army Corps of Engineers	Large hydroelectric storage is addressed in DBA-ES-HY-FC1. Run-of-river and pumped storage: deferred FC. Tidal: not yet scoped.	DBA-ES-HY-FC1
Solar	Utility-scale PV, solar thermal (CSP), distributed rooftop	FERC; NERC (interconnection); State PUC (distributed)	Utility-scale solar at cluster level: deferred FC. Distributed solar is addressed as an edge generation component in DBA-ES-DI-FC5.	Partial (FC5)
Wind	Onshore utility-scale, offshore, distributed	FERC; NERC; BOEM (offshore); State PUC	Wind at cluster level: deferred FC—distributed wind addressed as edge generation	Partial (FC5)

			component in DBA-ES-DI-FC5.	
Battery / BESS	Utility-scale BESS, distributed storage	FERC Order 841; NERC; State PUC	BESS at cluster level: deferred FC. Distributed BESS is addressed as the edge generation component in DBA-ES-DI-FC5.	Partial (FC5)
Geothermal, biomass, wave, tidal	Various	FERC; State PUC; applicable environmental agencies	Not yet scoped. Consequence chains have not yet been analyzed.	Not yet

2.3 TransCo — Transmission Owners and Operators

The transmission taxonomy covers the physical infrastructure connecting generation to load at bulk power system voltage levels. Voltage class specification and bus bar identifiers per circuit are the technical parameters that govern facility class placement within DBA-ES: the 345 kV threshold separates FC2 (EHV substations) from sub-transmission facilities outside the current series scope.

Infrastructure Element	Technical Specification	Regulatory Anchor	DBA-EN Series Treatment
EHV transmission substations	345–765 kV; metropolitan ring configuration; CIP-014 Tier 1/2 designation; LPT with RT_HVLLC ≥12 months	NERC CIP-014; FERC Order 693; TPL standards	DBA-ES-TX-FC2 (v0.1)
HVDC interconnects	>2,000 MW; cross-interconnection role; LCC or VSC technology; point-to-point (primary scope) or multi-terminal (deferred)	FERC; NERC FAC standards; applicable TPL	DBA-ES-TX-FC3 (v0.1)
Transmission lines	Voltage class per facility: 115 kV, 230 kV, 345 kV, 500 kV, 765 kV. Bus bar identifiers per circuit.	NERC FAC-001/002; FERC; State PUC	Deferred FC; addressed at the system level in TPL standards
SCADA and control communications	ICCP (Inter-Control Center Communications Protocol); SCADA RTUs; substation automation; protective relaying	NERC CIP-005/006/007; FERC Order 706	CE events in FC2, FC3; SL events in CntlCo instrument
Protective relay systems	Distance, differential, overcurrent, frequency, voltage protection; PRC-024 ride-through settings	NERC PRC standards; IEEE relay standards	CE-2/CE-3 events in FC2; CE-3 in FC5; UFLS position paper

2.4 DisCo — Distribution Functions

The distribution taxonomy covers the infrastructure delivering power from the transmission system to end-use customers. The PMU (Phasor Measurement Unit) is specifically called out in the DisCo control systems because it is the primary measurement instrument underlying the UFLS frequency wavefront analysis in the Predictive UFLS Ride-Through Position Paper — the cross-reference between DisCo instrumentation and the UFLS paper is a series linkage that must be maintained.

Infrastructure Element	Technical Specification	Regulatory Anchor	DBA-EN Series Treatment
Distribution substations with edge generation	HV/EHV input (115 kV+); 69–110 kV output; microgrid or DER assets; islanding capability with Command Broker architecture	NERC CIP (where applicable); IEEE 1547; State PUC; FERC Order 2222	DBA-ES-DI-FC5 (v0.1)
Loads	Commercial, industrial, residential. Load classification: firm/fixed vs. sheddable. Defense-critical loads: DSB finding (31 of 34 DoD most critical assets on commercial DisCo infrastructure).	State PUC; FERC (wholesale); DOD (defense-critical)	XC events in FC2, FC5; Tier 3 acceptance criteria
Distribution control systems	SCADA (2–5 second polling cycle); DCS; PLC; EMS (Energy Management System); PMU (Phasor Measurement Unit — wavefront analysis; cross-reference Predictive UFLS Ride-Through Position Paper)	NERC CIP; State PUC; IEEE standards	CE events in FC5; PMU data cited in UFLS position paper
Electricity markets	Wholesale market dispatch; retail rate structures; demand response programs; ancillary services	FERC (wholesale); State PUC (retail)	Market timing dependency in XC-4 events (FC3, FC4)

2.5 CntlCo — Balancing Authority, Reliability Coordinator, Transmission Operator

The CntlCo taxonomy is more tool-specific than the physical infrastructure taxonomies because the CntlCo's design basis is anchored in its software-data layer as much as its physical and human layers. The RC and BA tool suites, the NERC CIP compliance registry classification scheme, and the primary and backup control center communications architecture are all design basis elements addressed in the DBA-ES-CntlCo functional class instrument.

Function / Tool	Description	NERC Registration	DBA-EN Series Treatment
Reliability Coordinator (RC) tools	SCADA; Electronic Tagging System; Telephone System; OATI Suite of Applications; State Estimator; EPRI TagNet; Power World; RCIS (Reliability Coordinator Information System)	RC	SL events in DBA-ES-CntlCo §5; RC extension §7.2
Balancing Authority (BA) tools	Transmission Energy Scheduler; Eastern Interconnect Frequency Monitor; Security Constrained Unit Commitment; Telephone System; SCADA; RCIS; Energy Management System (EMS)	BA	SL events in DBA-ES-CntlCo §5; BA primary case §2
NERC CIP compliance registry	Critical cyber asset classification: BCA (BES Cyber Asset); EACMS (Electronic Access Control and Monitoring System); PACS (Physical Access Control System); PCA (Protected Cyber Asset). Impact classification: High / Medium / Low.	All CIP-registered entities	CE-4 (cyber intrusion) in FC2; SL-5 in CntlCo; governs CIP-014 designation trigger for FC2 scope
Primary and backup control center communications	Data communications; Voice; Radio; Telephone System; SCADA; ICCP (Inter-Control Center Communications Protocol). Primary and BCC must be geographically independent per DBA-ES-CntlCo acceptance criteria.	BA, RC, TOP	PL-4, PL-5 events in DBA-ES-CntlCo; BCC geographic independence acceptance criterion
Transmission Operator (TOP)	Operates transmission facilities in real time; implements RC and BA directives; controls switching at FC2 and FC3 facilities; must maintain manual field switching capability under SCADA loss	TOP	TOP extension in DBA-ES-CntlCo §7.3

2.6 Oil and Natural Gas

The Oil and Natural Gas branch of the energy taxonomy covers two commodity streams with partially overlapping infrastructure. The governing cross-sector cascade — the gas-electric bidirectional feedback loop — is addressed in Section 3. The taxonomy below covers the upstream, midstream, downstream, and storage segments of each commodity stream.

Segment	Infrastructure	Technical / Operational Notes	Primary Regulators	DBA-EN Series Treatment
Crude Oil — Upstream	Onshore fields; offshore fields; gathering systems	Wellhead pressure management, produced water handling, and gathering line integrity	PHMSA; EPA; State environmental; BSEE (offshore)	Not yet scoped in DBA-OIL series

Crude Oil — Midstream	Terminals; pipelines; storage (tank farms)	Crude transport by pipeline, rail, and marine. Tank farm fire and explosion risk. Colonial Pipeline paradigm applicable.	PHMSA; TSA; FERC; EPA	DBA-OIL-FC1 (v1.1) — petroleum refining; midstream FC deferred
Petroleum Processing	Refineries, blending terminals, petrochemical plants	Process unit fire and explosion as the governing SE event. Long lead time for major process units (reactors, distillation columns). RT_HVLLC concept applicable.	TSA; EPA; OSHA; State environmental	DBA-OIL-FC1 (v1.1)
Natural Gas — Upstream	Onshore fields; offshore fields; gathering systems; processing plants	Gas processing separates NGLs from methane. Processing plant disruption reduces the pipeline quality gas supply.	PHMSA; FERC; EPA; BSEE (offshore)	Not yet scoped in DBA-GAS series
Natural Gas — Transmission	Interstate and intrastate pipelines; compressor stations; metering and regulation (M&R) stations	Compressor stations in critical GenCo supply corridors are the governing FC4 facility class. GSD metric governs electric consequence.	FERC; PHMSA; TSA (security directives)	DBA-GAS-FC1 (v1.5); DBA-ES-GC-FC4 (v0.2)
Natural Gas — Distribution	Local distribution companies (LDCs), service laterals, and customer meters	Residential heating dependency: dual-use pipeline (generation fuel + heating fuel) produces compound XC-4 consequence during winter peak.	State PUC; PHMSA	XC-4 in DBA-ES-GC-FC4; not yet addressed in DBA-GAS series
LNG Facilities	Liquefaction terminals; storage tanks; regasification terminals; import/export terminals	LNG as a backup supply for pipeline disruption. Sabotage of LNG import terminal removes backup gas supply during extended pipeline outage.	FERC; PHMSA; USCG; DOE (export)	DBA-GAS-UC-LNG (v1.1)
Storage	Underground storage (depleted reservoirs, aquifer, salt cavern); above-ground LNG storage	Strategic storage buffer: days to weeks of deliverability. Storage disruption extends GSD duration beyond compressor station recovery.	FERC; PHMSA; State	Addressed as PE context in DBA-GAS-FC1; dedicated FC deferred

Section 3 — The Gas-Electric Cascade: Cross-Sub-Sector Postulated Event

3.1 Status as a First-Class Postulated Event

The gas-electric bidirectional cascade is established in this document as a first-class cross-sub-sector postulated event at the DBA-EN level. It is not a secondary consequence or a beyond-design-basis contingency: it is a primary postulated event that must be analyzed in both the DBA-ES series (from the electric system's perspective) and the DBA-GAS series (from the gas system's perspective). The two analyses are complementary; neither substitutes for the other.

The bidirectional character of the cascade — gas disruption reduces electric generation; electric disruption reduces gas compression — was established in DBA-ES-Sector_Framework §3.5 and is carried through every relevant FC instrument. Winter Storm Uri (February 2021) demonstrated both directions simultaneously and empirically. An adversary who understands the feedback structure can initiate the cascade from whichever end is more accessible and rely on the bidirectional dependency to amplify the consequence without further action.

3.2 Cascade Analysis Table

Cascade Direction	Mechanism	Governing Metric	Primary DBA-EN Instrument
Gas → Electricity	Natural gas compressor station loss reduces pipeline delivery pressure below the GenCo minimum inlet threshold. Gas-fired generation curtails output or trips offline. The electric system loses generation capacity. Approximately 40% of US electric generation is gas-fired; gas transmission disruption in a critical corridor reduces generation capacity at the electric system level.	GSD (Generation Supply Duration): continuous period of full gas-fired generation curtailment due to insufficient pipeline pressure. Defined in DBA-ES-GC-FC4 §1.3.	DBA-ES-GC-FC4; DBA-GAS-FC1
Electricity → Gas	Electric supply loss to electrically driven compressor stations reduces compression capacity. Pipeline pressure in the supply corridor declines. Gas supply to gas-fired generation and heating customers is curtailed. Self-reinforcing: reduced gas supply further reduces gas-fired generation, further reducing electric supply to compressor station auxiliaries.	GSD (reverse cascade): same metric applied to the electrically-initiated cascade. TCI (Transfer Capability Isolation, from DBA-ES-TX-FC3) may apply if HVDC or transmission loss initiates the electric supply disruption.	DBA-ES-GC-FC4 (XC-2); DBA-ES-TX-FC2; DBA-ES-TX-FC3
Compound (bidirectional)	Winter Storm Uri (February 2021) demonstrated the full bidirectional cascade: extreme cold simultaneously	Compound GSD + load curtailment. XC-4 event in DBA-ES-GC-FC4:	DBA-ES-GC-FC4 (XC-4); cross-

	increased electric demand and reduced gas-fired generation supply by disabling compressor stations and gas wells. The compound event eliminated both directions of the cascade mitigation simultaneously. An adversary can replicate this compound cascade deliberately through coordinated timing.	winter peak demand timing maximizes both heating fuel and electric generation consequences.	reference DBA-GAS-FC1

3.3 Cross-Sub-Sector Coordination Requirement

The gas-electric cascade is the primary reason DBA-EN-SF exists as a document distinct from the sub-sector frameworks. Each sub-sector framework addresses the cascade from its own consequence perspective: DBA-ES instruments analyze the electric consequence of gas infrastructure failure; DBA-GAS instruments analyze the gas infrastructure consequence of electric supply failure. The DBA-EN-SF level is where the full bidirectional cascade is formally established as a postulated event category that both sub-sector analyses must address.

Specifically, DBA-ES-GC-FC4 and DBA-GAS-FC1 have a required scope boundary coordination (DBA-ES-GC-FC4 Deferred Item DI-3). The same physical compressor station is within the scope of both instruments, but is analyzed from different consequence perspectives. The coordination protocol establishing which instrument governs for which consequence and how the two analyses relate must be completed before either instrument reaches v0.2. DBA-EN-SF is the authority document under which that coordination protocol will be established.

3.4 Cross-Sector Consequence Elevation Rule

The cross-sector consequence elevation rule is established here as a governing architectural requirement for all DBA-EN series instruments. Its purpose is to ensure that the design basis of an energy sector asset reflects not only the asset's own consequence footprint within the energy sector, but the consequence footprint of the facilities in other sectors that depend on it for their primary energy supply.

The NERC gap this rule addresses. The North American Electric Reliability Corporation's CIP standards assign impact levels to bulk electric system cyber assets based on the electric-sector consequences of their loss — the potential for widespread instability, cascading failures, or load shedding within the interconnection. NERC CIP does not require, and has not established, any mechanism by which the consequence designation of a facility in another sector elevates the design basis or impact classification of the energy sector asset that supplies it. A hospital system, water treatment plant, nuclear facility, or military installation that depends on a single electric substation for its primary power supply does not, under current NERC standards, cause that substation's design basis to reflect the hospital's, plant's, reactor's, or installation's consequence tier. The DBA-EN-SF cross-sector consequence elevation rule fills this gap as a pre-regulatory design basis requirement. It does not modify or supersede NERC CIP standards; it establishes the design basis requirements that apply to energy sector facilities within the DBA-EN analytical framework independently of NERC regulatory status.

The rule. A DBA-EN series energy sector asset whose loss would deprive a facility carrying a Tier 2 or Tier 3 consequence designation in any DBA series instrument of its primary energy supply must carry tightened acceptance criteria for that supply relationship in its own DBA-EN design basis. Tightening means the acceptance criteria governing the supply path — restoration timelines, backup posture duration requirements, redundancy specifications, and denial-condition maintainability — must be set to a standard sufficient to ensure that the dependent facility's consequence tier is not exceeded through energy supply failure alone. This rule does not elevate the threat tier assigned to the energy sector asset; only the acceptance criteria for the specific supply relationship tighten.

Trigger condition. The rule fires when two conditions are simultaneously satisfied. First, the dependent facility carries a Tier 2 or Tier 3 consequence designation in a DBA series instrument — meaning it has been analyzed within the DBA methodology and its loss has been determined to produce grid-level or cross-sector consequences. Second, the energy sector asset is the sole necessary supply path: the dependent facility does not have an alternate energy supply of equivalent capacity that would maintain its operations if the DBA-EN asset were unavailable. This single-path dependency criterion is drawn directly from the FC4 facility class definition (DBA-ES-GC-FC4 §1.2), where it governs scope inclusion for compressor stations in GenCo supply corridors. Applied here at the DBA-EN-SF level, it governs consequence elevation across all energy sector facility classes and all dependent sectors. A dependent facility with a firm alternate energy supply of equivalent capacity does not trigger the elevation; the redundancy is itself the design basis mitigation, and the acceptance criteria for the individual supply path need not be tightened beyond the facility's own baseline.

Self-referential trigger and series growth. The trigger is self-referential within the DBA series: the Tier 2 or Tier 3 designation of the dependent facility is established by its own DBA instrument, not by any external regulatory designation. This design is intentional. It means the elevation rule does not depend on external designation systems — NERC CIP, DHS critical infrastructure designations, DoD defense-critical lists — that change outside the series' control and that do not currently incorporate cross-sector consequence logic. It also means the rule automatically expands as the DBA series matures. When the DBA-W series assigns a Tier 3 designation to a water treatment facility, every DBA-EN energy asset that is the sole supply path for that facility is immediately subject to the elevation rule, without any amendment to this document. The rule is stated once, at the DBA-EN-SF level, and propagates forward through the series' own growth.

Direction and future state. The rule, as currently stated, is unidirectional: the energy sector carries the design basis obligation imposed by dependent sectors' consequence tiers. The full bidirectional application — in which a DBA-W or DBA-MA nuclear facility's design basis is also elevated by the consequence tier of the energy infrastructure it depends on — is the intended future state of the rule as those sector series develop their own consequence designations and their authors are positioned to incorporate the reciprocal obligation. The European EPCIP / NIS2 framework, which designates a facility as critical if it supplies power to a critical element in another sector irrespective of cascade potential, is the governing international analog. The current unidirectional implementation already extends design basis requirements onto the energy sector beyond those of the NERC CIP standards; the full bilateral implementation is deferred to the maturation of the dependent sector series.

Application to the FC4 / GAS-FC1 scope boundary. The cross-sector consequence elevation rule, together with the consequence-perspective separation rule and the acceptance criteria assignment rule established in §3.3, constitutes the complete coordination protocol for DBA-ES-GC-FC4 and DBA-GAS-FC1 at dual-scope facilities. A compressor station in a critical GenCo supply corridor carries a design basis under DBA-ES-GC-FC4 that is governed by the electric sector consequences of its loss (GSD metric; GenCo supply corridor acceptance criteria). That same station carries a design basis under DBA-GAS-FC1 that is governed by the gas infrastructure consequences of its loss (gas sector consequence metrics). Under the cross-sector consequence elevation rule, the DBA-ES-GC-FC4 acceptance criteria for the station must also reflect any Tier 2 or Tier 3 DBA consequence designation carried by the gas-fired generation facilities it serves in the supply corridor — which they carry by definition, since GenCo generation curtailment is a Tier 2 consequence in the DBA-ES taxonomy. This means FC4 acceptance criteria are subject to the elevation rule at every facility within the class: the rule confirms and formalizes the consequence-based design basis already established in DBA-ES-GC-FC4 §7, and extends it explicitly to any other DBA-EN energy asset that is the sole supply path for a Tier 2 or Tier 3 DBA-designated facility in any sector. This section resolves Deferred Item DI-3 of DBA-ES-GC-FC4. Both instruments may proceed to v0.2 development.

Section 4 — Series-Wide Architectural Requirements

4.1 DB→DBA→DBA-MA Three-Layer Progression

All instruments in the Energy sector series — sub-sector frameworks, facility-class instruments, functional class instruments, and use case documents — apply the DB→DBA→DBA-MA three-layer progression. The progression is non-negotiable: a facility-class instrument that addresses only the DBA layer without explicitly establishing the DB layer, or that addresses the DBA-MA extension without the DBA envelope, is architecturally incomplete.

Non-nuclear sector frameworks and facility-class instruments cannot assume reader familiarity with the DB→DBA→DBA-MA architecture and must state it explicitly. This requirement was established in DBA-ES-Sector_Framework and is adopted here as an Energy-sector-wide requirement.

4.2 Dual Design Basis Architecture

Consistent with DBA-ES §3.4 and the AI Governance series (Project 2), any energy sector facility may simultaneously carry a sector-specific physical Design Basis established by the applicable DBA-EN series instrument and an AI Governance Design Basis governing the deployment of ML and Gen AI systems in the facility's operational and control environment. The two design bases interact at the DBA level and must be developed in coordination.

Each sub-sector framework and facility-class instrument must acknowledge the dual-DB architecture and identify the interaction points between the physical DBA and the AI Governance DBA specific to its facility type. This requirement applies to all new instruments and to existing instruments on their next substantive revision pass.

4.3 Bright Line and Command Broker Architecture

The Bright Line and Command Broker architecture is a series-wide requirement for all energy sector instruments. The Bright Line defines the boundary between autonomous ML system operation (deterministic, formally testable, autonomous operation permitted below the Bright Line) and human Command Broker oversight (non-deterministic Gen AI systems require a human Command Broker above the Bright Line).

The term "Bright Line" is used exclusively in the ML/AI governance sense throughout all DBA-EN series instruments. Any other use of the term in an energy sector instrument — for example, to mean federal coordination triggers or escalation thresholds — must be replaced with a distinct term. The applicable replacement term for federal coordination thresholds is "Federal Coordination Threshold."

4.4 Consequence Metrics

Three consequence-duration metrics are established across the Energy sector series. Each is a series standard and must be used consistently in any instrument that addresses the relevant consequence type.

RT_HVLLC (Recovery Timeline for High-Value Long-Lead Components): the time required to restore a facility to service following destruction of its most difficult-to-replace major component, expressed in months. Established as a series standard in DBA-ES-Sector_Framework §6.2. Originated in GenMix Optimization Whitepaper (Roxey, Eclectic Technologies, 2026). Applies to all FC instruments where long-lead-time equipment is the governing consequence element.

TCI (Transfer Capability Isolation): the continuous duration during which cross-interconnection transfer capability is unavailable following loss or block of an HVDC link, expressed in hours, days, or months. Established in DBA-ES-TX-FC3 §1.3. Applies to HVDC facility class instruments.

GSD (Generation Supply Duration): the continuous period during which gas-fired generation capacity in an affected GenCo supply corridor is curtailed to zero due to insufficient pipeline delivery pressure, expressed in hours or days. Established in DBA-ES-GC-FC4 §1.3. Applies to gas compressor stations and gas transmission instruments where the governing consequence chain runs through the gas-electric cascade.

4.5 Facility Class Naming Convention

The naming convention for Energy sector facility class instruments is: DBA-[SUB-SECTOR]-[TYPE CODE]-FC[NUMBER]. Type codes are assigned to distinguish facility classes within the same sub-sector by their governing infrastructure type. The DBA-DC series retains its existing designation without modification.

Designation	Sub-Sector	Type Code	Facility / Functional Class
DBA-EN-SF	Energy (parent)	SF = Sector Framework	This document — Energy sector parent
DBA-ES-Sector_Framework	Electric	SF = Sector Framework	Electric sub-sector parent
DBA-ES-HY-FC1	Electric	HY = Hydroelectric	Large hydroelectric storage (>500 MW)
DBA-ES-TX-FC2	Electric	TX = Transmission	Large EHV transmission substation (345–500 kV)
DBA-ES-TX-FC3	Electric	TX = Transmission	HVDC interconnect (>2,000 MW)
DBA-ES-GC-FC4	Electric	GC = Gas-electric Cascade	Major natural gas compressor station (GenCo supply corridor)
DBA-ES-DI-FC5	Electric	DI = Distribution	Distribution substation with edge generation
DBA-ES-CntlCo	Electric	CntlCo = functional class	BA / RC / TOP functional class instrument
DBA-DC (unchanged)	Electric / IT	DC = Data Center	Hyperscale data center (>100 MW IT load)
DBA-GAS-Sector_Framework	Natural Gas	SF = Sector Framework	Natural gas sub-sector parent
DBA-GAS-FC1	Natural Gas	—	Natural gas transmission
DBA-OIL-Sector_Framework	Petroleum	SF = Sector Framework	Petroleum sub-sector parent
DBA-OIL-FC1	Petroleum	—	Petroleum refining

Section 5 — Cross-Sector Dependencies Beyond Gas-Electric

5.1 Water Sector Dependency

Water treatment and distribution facilities are significant electric loads and, in some configurations, significant gas consumers (for backup generation and process heat). Loss of electric supply to water sector facilities is a Tier 3 consequence in the DBA-ES series. The DBA-W (Water sector) series addresses this dependency from the water sector's perspective. DBA-EN-SF does not govern DBA-W instruments; the cross-reference is noted here to establish the boundary.

5.2 Data Center Dependency

Hyperscale data centers are addressed in the DBA-DC series, which is an Electric sector sub-series instrument. Data centers are electricity consumers — not electricity infrastructure — per NIPP taxonomy. Their classification in the Electric sector design basis series reflects their consequence chain, not their sector membership. The WM-3 workload migration cross-interconnection load transient is the governing cross-sector cascade event for the data center class and is addressed in DBA-DC-L1 and DBA-ES-Sector_Framework.

5.3 Nuclear Sector Grid Dependency

Nuclear generating facilities are significant contributors to the bulk electric system. The Zaporizhzhia scenario demonstrated that military action against a nuclear facility is simultaneously a nuclear safety event and a grid reliability event. The DBA-MA nuclear series addresses the nuclear safety consequences; the DBA-ES series addresses the grid reliability consequences of nuclear facility loss. The boundary is defined in the DBA-MA Introduction §6.3: facility consequence analysis lives in DBA-MA; grid consequence analysis lives in DBA-ES. DBA-EN-SF is the appropriate level at which this boundary is formally established. The NERC regulatory instrument governing the interface between nuclear plant operations and the bulk electric system is NUC-001-4 (Nuclear Plant Interface Coordination). NUC-001-4 requires Nuclear Plant Generator Operators and Transmission Entities to develop and maintain Nuclear Plant Interface Requirements (NPIRs) — agreed operating limits that ensure nuclear plant safe operation and shutdown can be achieved within the constraints of the transmission system, and that the transmission system can be operated within the constraints of the nuclear plant's licensing requirements. NUC-001-4's NPIR coordination mechanism is the existing regulatory pathway through which nuclear-facility-specific grid reliability requirements are made binding on Transmission Operators, Balancing Authorities, Reliability Coordinators, and other Transmission Entities. The DBA-EN series does not replicate this mechanism; it establishes the design basis requirements that apply to grid infrastructure when nuclear facility loss is postulated as a consequence event. The two frameworks are complementary: NUC-001-4 governs the operational interface during normal and emergency conditions; the DBA-EN design basis governs the physical and consequence envelope that the electric sector infrastructure must demonstrate it is capable of withstanding when a nuclear facility is lost to the grid as a generation resource.

Section 6 — Development Status and Open Items

6.1 Current Development Status

The Electric sub-sector series (DBA-ES) is the most developed of the three sub-sector series. DBA-ES-Sector_Framework is at v0.6 with six active FC instruments and a defined CntlCo functional class instrument, all at v0.1 DB-layer status. The DBA and MA extension layers for FC5 and CntlCo are deferred to v0.2 of those instruments. The Electric sector series is ready to anchor the new Energy Sector project.

The Natural Gas sub-sector series (DBA-GAS) is at v1.4 for its sector framework with one active FC instrument (DBA-GAS-FC1 at v1.3), one use case (DBA-GAS-UC-LNG at v1.1), and one policy companion (DBA-GAS-POL_Deterrence at v1.2). The series is substantively developed at the transmission facility class level.

The Petroleum sub-sector series (DBA-OIL) is at v1.3 for its sector framework with one active FC instrument (DBA-OIL-FC1 at v1.1) covering petroleum refining. Midstream and upstream petroleum facility classes are not yet scoped.

6.2 Open Items

OI-1: CLOSED. DBA-ES-GC-FC4 / DBA-GAS-FC1 scope boundary coordination. Resolved by §3.4 (Cross-Sector Consequence Elevation Rule) of this document. The consequence-perspective separation rule, the acceptance criteria assignment rule, and the cross-sector consequence elevation rule together constitute the complete coordination protocol for dual-scope facilities. Both instruments may proceed to v0.2 development.

OI-2: GenCo facility classes for coal, oil-fired, and renewable generation types. The GenCo taxonomy in Section 2.2 identifies fossil fuel (non-gas) and renewable generation types as deferred facility classes. The scoping decision — whether these warrant dedicated FC instruments or are addressed through consequence-chain analysis in existing instruments — should be made before DBA-EN-SF reaches v0.3.

OI-3: Upstream and midstream petroleum facility classes. DBA-OIL-FC1 covers petroleum refining. Crude oil upstream (production fields, gathering) and midstream (pipelines, terminals, tank farms) are not yet scoped. The Colonial Pipeline ransomware event is the primary evidentiary anchor for a midstream petroleum FC instrument.

OI-4: Natural gas distribution and storage facility classes. The local distribution company (LDC) and underground storage segments of the natural gas taxonomy are not yet addressed in the

DBA-GAS series. The residential heating dependency (XC-4 in DBA-ES-GC-FC4) requires a DBA-GAS distribution instrument to be fully addressed from the gas sector perspective.

OI-5: DBA-EN-SF / DBA-W boundary. The Water sector dependency is noted in Section 5.1, but the formal boundary between DBA-EN-SF governance and DBA-W governance is not yet established. This boundary should be addressed at the next substantive revision of DBA-EN-SF.

Section 7 — References

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Roxey, Tim. Design Basis Energy Sector Mindmap—version 2. Eclectic Technologies, June 2026. (Source document for Section 2 taxonomy.)

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